Project

Business Case:

Telco is a telecommunication company based in California and operating in over 1000 cities . Recently, Telco has been concerned about their customer churn rate and wants to be able to predict what the future customer churn rate would be, using statistical models to understand and improve their customer retention strategy.

They have provided me with a record of 7k rows of customer data to work with.

About the data:

- Customers who left within the last month – Churn
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges.
- Demographic info about customers – gender, age range, and if they have partners and dependents.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
Customerl	Count	Country	State	City	Zip Code	Lat Long	Latitude	Longitude	Gender	Senior Citi	Partner	Dependen	Tenure Mo	Phone Ser	Multiple Li	Internet Se	Online Sec	Online Bac
3668-QPYI	1	United Stat	California	Los Angele	90003	33.964131	33.96413	-118.273	Male	No	No	No	2	Yes	No	DSL	Yes	Yes
9237-HQIT	1	United Stat	California	Los Angele	90005	34.059281	34.05928	-118.307	Female	No	No	Yes	2	Yes	No	Fiber optic	No	No
9305-CDSI	1	United Stat	California	Los Angele	90006	34.048013	34.04801	-118.294	Female	No	No	Yes	8	Yes	Yes	Fiber optic	No	No
7892-POO	1	United Stat	California	Los Angele	90010	34.062125	34.06213	-118.316	Female	No	Yes	Yes	28	Yes	Yes	Fiber optic	No	No
0280-XJGE	1	United Stat	California	Los Angele	90015	34.039224	34.03922	-118.266	Male	No	No	Yes	49	Yes	Yes	Fiber optic	No	Yes
4190-MFLI	1	United Stat	California	Los Angele	90020	34.066367	34.06637	-118.31	Female	No	Yes	No	10	Yes	No	DSL	No	No
8779-QRD	1	United Stat	California	Los Angele	90022	34.02381,	34.02381	-118.157	Male	Yes	No	No	1	No	No phone s	DSL	No	No
1066-JKSG	1	United Stat	California	Los Angele	90024	34.066303	34.0663	-118.435	Male	No	No	No	1	Yes	No	No	No interne	No interne
6467-CHF	1	United Stat	California	Los Angele	90028	34.099869	34.09987	-118.327	Male	No	Yes	Yes	47	Yes	Yes	Fiber optic	No	Yes
8665-UTDI	1	United Stat	California	Los Angele	90029	34.089953	34.08995	-118.295	Male	No	Yes	No	1	No	No phone s	DSL	No	Yes
8773-HHU	1	United Stat	California	Los Angele	90032	34.078821	34.07882	-118.178	Female	No	No	No	17	Yes	No	DSL	No	No
6047-YHP	1	United Stat	California	Los Angele	90039	34.110845	34.11085	-118.26	Male	No	No	Yes	5	Yes	No	Fiber optic	No	No
5380-WJK	1	United Stat	California	Los Angele	90041	34.137412	34.13741	-118.208	Male	No	No	Yes	34	Yes	Yes	Fiber optic	No	Yes
8168-UQW	1	United Stat	California	Los Angele	90042	34.11572,	34.11572	-118.193	Female	No	No	Yes	11	Yes	Yes	Fiber optic	No	No
7760-OYPI	1	United Stat	California	Los Angele	90056	33.987945	33.98795	-118.37	Female	No	No	Yes	2	Yes	No	Fiber optic	No	No
9420-LOJK	1	United Stat	California	Los Angele	90061	33.92128,	33.92128	-118.274	Female	No	No	Yes	15	Yes	No	Fiber optic	Yes	Yes
7495-OOK	1	United Stat	California	Los Angele	90063	34.044271	34.04427	-118.185	Female	Yes	Yes	Yes	8	Yes	Yes	Fiber optic	No	Yes
1658-BYG	1	United Stat	California	Los Angele	90065	34.108833	34.10883	-118.23	Male	Yes	No	Yes	18	Yes	Yes	Fiber optic	No	No
5698-BQJ	1	United Stat	California	Beverly Hil	90211	34.063947	34.06395	-118.383	Female	No	No	Yes	9	Yes	Yes	Fiber optic	No	No
5919-TMR	1	United Stat	California	Huntingtor	90255	33.97803,	33.97803	-118.217	Female	No	No	Yes	1	Yes	No	Fiber optic	No	No
9191-MYQ	1	United Stat	California	Lynwood	90262	33.923573	33.92357	-118.201	Female	No	Yes	Yes	7	Yes	No	Fiber optic	No	No
8637-XJIV	1	United Stat	California	Marina Del	90292	33.977468	33.97747	-118.445	Female	No	No	Yes	12	Yes	Yes	Fiber optic	Yes	No
0278-YXO	1	United Stat	California	Inglewood	90301	33.956445	33.95645	-118.359	Male	No	No	No	5	Yes	No	No	No interne	No interne
4598-XLKN	1	United Stat	California	Inglewood	90303	33.936291	33.93629	-118.333	Female	Yes	Yes	Yes	25	Yes	No	Fiber optic	No	Yes
3192-NQE	1	United Stat	California	Santa Mon	90403	34.031529	34.03153	-118.491	Male	No	Yes	No	68	Yes	Yes	Fiber optic	No	Yes
0486-HEC	1	United Stat	California	Torrance	90504	33.867257	33.86726	-118.331	Male	No	Yes	Yes	55	Yes	Yes	Fiber optic	Yes	Yes
4846-WHA	1	United Stat	California	Whittier	90602	33.972119	33.97212	-118.02	Female	Yes	Yes	Yes	37	Yes	Yes	Fiber optic	No	No
5299-RULC	1	United Stat	California	Whittier	90606	33.976678	33.97668	-118.066	Female	No	Yes	Yes	10	Yes	Yes	Fiber optic	Yes	No
0404-SWR	1	United Stat	California	La Habra	90631	33.940619	33.94062	-117.951	Male	No	No	Yes	3	Yes	Yes	Fiber optic	No	No

Fig 2.1 - Customer data spreadsheet for Telco containing - 7k customer records.

Step 1 - Data Preprocessing

- **Data cleaning:** Duplicates were checked and cleared out
- Null values were deleted
- The data types were adjusted to the right data types
- Checks were made for outliers

Data streamlining:

The data was loaded into the python library to help with the analysis to be conducted.

Predicting Telco Customer Churn

```
#Importing the Pandas and Numpy Library
import pandas as pd
import numpy as np
```

```
#Reading the data
df = pd.read_csv(r"C:\Users\timil\OneDrive\Documents\Java testing\Telco_customer_churn.csv")
```

```
df.columns
```

```
Index(['CustomerID', 'Count', 'Country', 'State', 'City', 'Zip Code',
       'Lat Long', 'Latitude', 'Longitude', 'Gender', 'Senior Citizen',
       'Partner', 'Dependents', 'Tenure Months', 'Phone Service',
       'Multiple Lines', 'Internet Service', 'Online Security',
       'Online Backup', 'Device Protection', 'Tech Support', 'Streaming TV',
       'Streaming Movies', 'Contract', 'Paperless Billing', 'Payment Method',
       'Monthly Charges', 'Total Charges', 'Churn Label', 'Churn Value',
       'Churn Score', 'CLTV', 'Churn Reason'],
      dtype='object')
```

Fig 2.2 - Database containing table aforementioned data

Exploratory Data Analysis and Visualization

Exploratory Data Analysis and Visualization

```
df.describe().transpose()
```

	count	mean	std	min	25%	50%	75%	max
Count	7043.0	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.000000
Zip Code	7043.0	93521.964646	1865.794555	90001.000000	92102.000000	93552.000000	95351.000000	96161.000000
Latitude	7043.0	36.282441	2.455723	32.555828	34.030915	36.391777	38.224869	41.962127
Longitude	7043.0	-119.798880	2.157889	-124.301372	-121.815412	-119.730885	-118.043237	-114.192901
Tenure Months	7043.0	32.371149	24.559481	0.000000	9.000000	29.000000	55.000000	72.000000
Monthly Charges	7043.0	64.761692	30.090047	18.250000	35.500000	70.350000	89.850000	118.750000
Churn Value	7043.0	0.265370	0.441561	0.000000	0.000000	0.000000	1.000000	1.000000
Churn Score	7043.0	58.699418	21.525131	5.000000	40.000000	61.000000	75.000000	100.000000
CLTV	7043.0	4400.295755	1183.057152	2003.000000	3469.000000	4527.000000	5380.500000	6500.000000

Fig 2.3 - Table showing statistical measure of numerical columns

CHURN column tells us about the number of Customers who left within the last month.

- Around 26% of customers left the platform within the last month.

```
plt.figure(
figsize = (20,7))
plt.title ('Churn Label')
sns.countplot(x ='Churn Label', data = df, palette = 'YlGnBu')
for i, v in enumerate(plt.gca().patches):
    plt.gca().text(v.get_x()+v.get_width()/2,v.get_height(), str(round(v.get_height(),2)), ha = 'center', va = 'bottom')
plt.show()
```

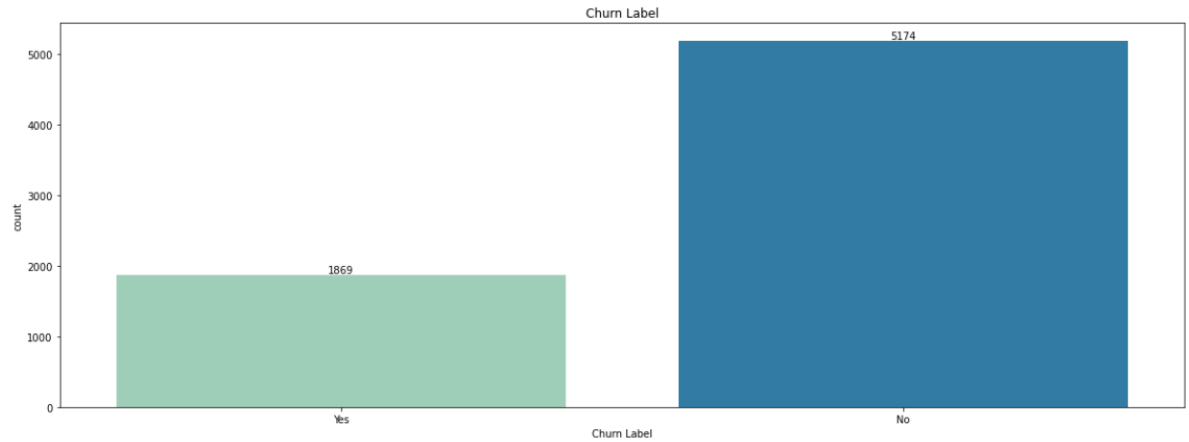

Fig 2.4 - Bar plot of Churn Label

- **Gender** - The churn percent is almost equal in both Male and Female
- The percentage of churns is lower in the case of **senior citizens**
- Customers with **Dependents** have lower churn rate as compared to those who don't have Dependents.
- Customers who are on a monthly subscription have a higher churn rate than customers who pay annually and those on a 2-year subscription.

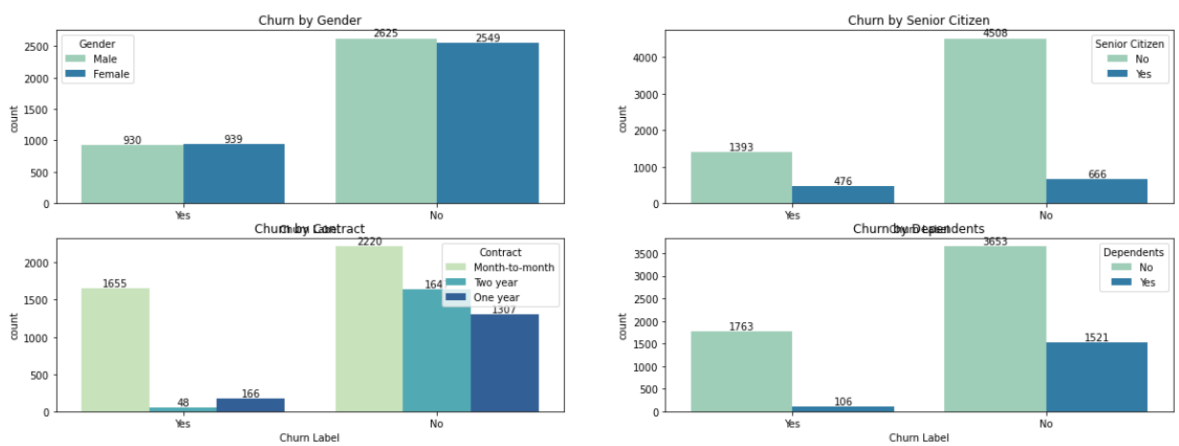

Fig 2.5 - Bar plots of Churn label per Gender/Senior Citizen/Contract/Dependent

The distribution shows that the data is left skewed, and the largest frequency of churn score is between 65 to 80.

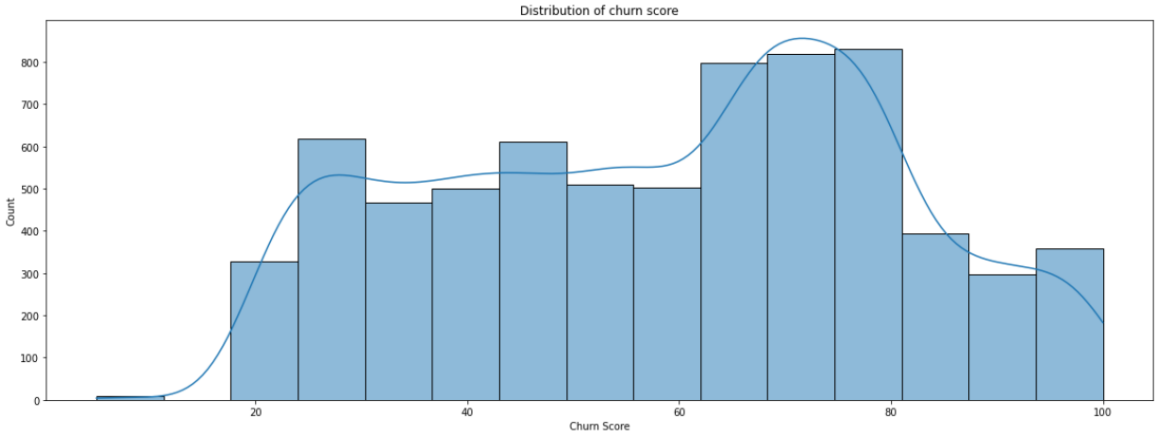

Fig 2.6 - Distribution of Churn score

The distribution shows that the data is left skewed, and the largest frequency of customer lifetime value is between 4,500 to 5,500.

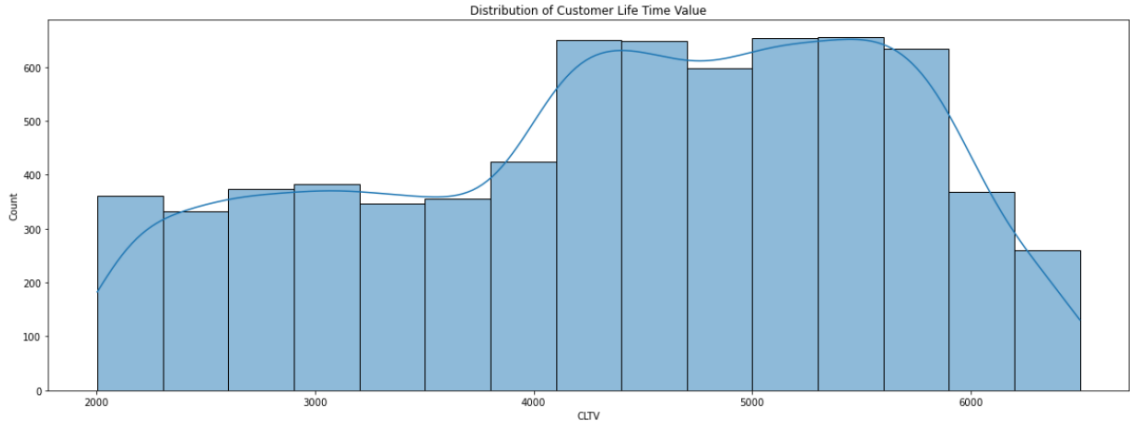

Fig 2.7 - Distribution of Customer Lifetime Value

Tenure: The median tenure for customers who have left is around 10 months.

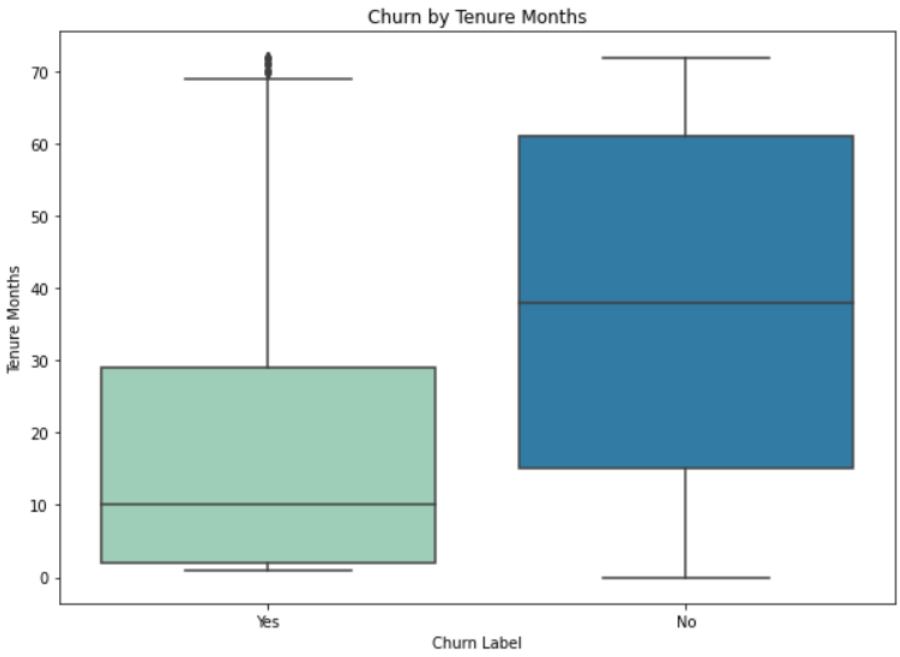

Fig 2.8 - Churn by Tenure Months

Monthly Charges: Customers who have churned have high monthly charges. The median is above 75.

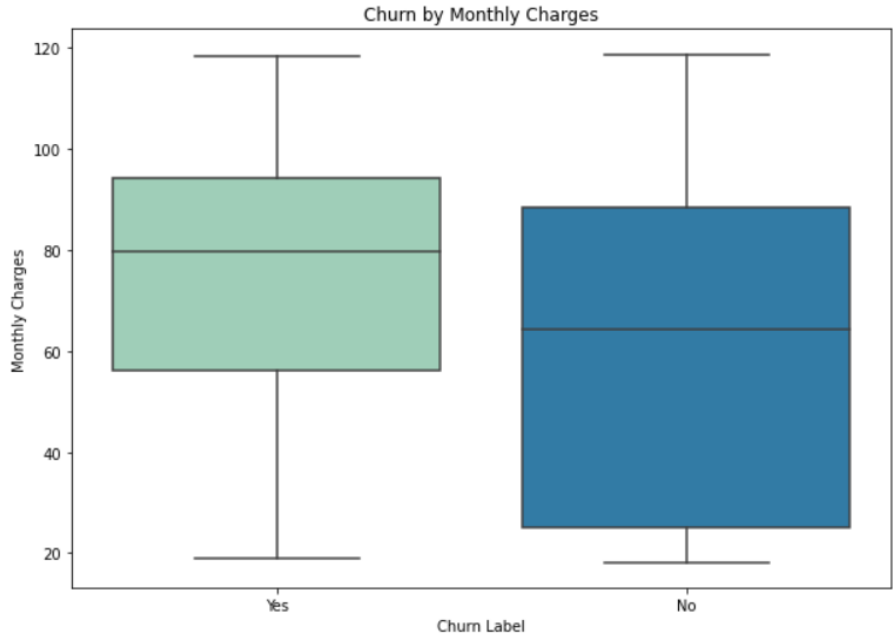

Fig 2.9 - Churn by Monthly Charges

Data Modeling

- **Feature selection:** Domain knowledge and correlation function were used for feature engineering
- Applying **oversampling strategy** to cater for imbalanced data
- Transforming **categorical data** into numeric for machine learning
- Running RandomForestClassifier model

Feature selection is the process of reducing the number of input variables when developing a predictive model. The features selected were done using domain knowledge and correlation function. Gender, Contract, Payment Method, Monthly Charges, Total, Charges, Customer Lifetime value were all the features selected to predict the Churn Value.

Feature Selection

```
In [20]: X = df[['Gender', 'Contract', 'Payment Method', 'Monthly Charges', 'Total Charges','CLTV' ]]
         y = df['Churn Value']

In [21]: df['Churn Value'].unique()

Out[21]: array([1, 0], dtype=int64)

In [22]: df['Churn Value'].value_counts().plot(
         kind = 'bar'
         )

Out[22]: <AxesSubplot:>
```

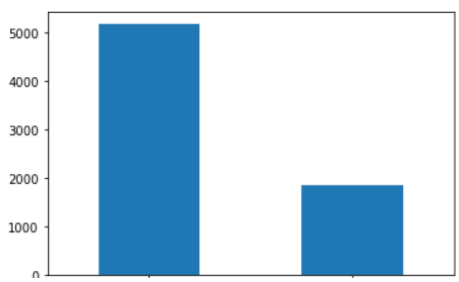

Fig 3.0 - Feature selection diagram

Oversampling is a data augmentation technique utilized to address class imbalance problems in which one class significantly outnumbers the other to have a more balanced data of the minority class.

click to expand output; double click to hide output

```
In [28]: # Normalizing the data
         from imblearn.over_sampling import SMOTE

In [29]: sm = SMOTE(sampling_strategy = 'auto', random_state = 42)
         X_resampled, y_resampled = sm.fit_resample(x_reshaped,y)

In [30]: y_resampled.value_counts().plot(
         kind = 'bar');
```

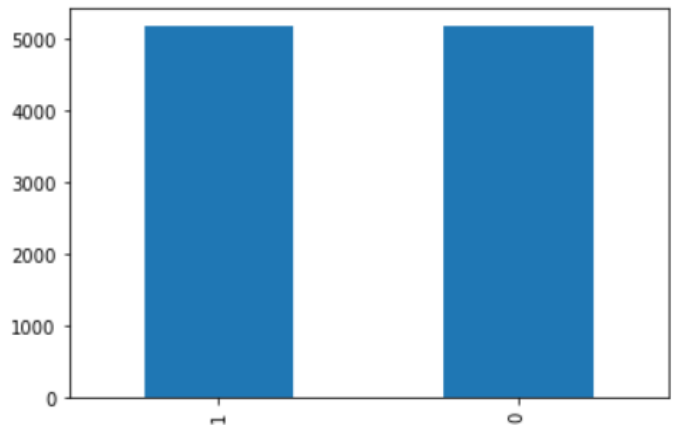

Fig 3.1 - Diagram showing resampled data

Categorical data from feature selection are converted to numerical features using scikit-learn column transformer library with OneHotEncoder. This transforms non-numerical features to numerical features for machine learning models. It is stored in a variable called **x_reshaped**.

Transforming categorical data to numeric

```
[15]: from sklearn.preprocessing import OneHotEncoder
      from sklearn.compose import ColumnTransformer

[16]: ct = ColumnTransformer(transformers = [('trans', OneHotEncoder(sparse_output = False),['Gender','Contract','Payment Method','Mont
      ct

[16]: ▸ ColumnTransformer
        ▸ trans          ▸ remainder
          ▸ OneHotEncoder  ▸ passthrough

[17]: x_reshaped = ct.fit_transform(X)
      x_reshaped

[17]: array([[0., 1., 1., ..., 0., 0., 0.],
             [1., 0., 1., ..., 0., 0., 0.],
             [1., 0., 1., ..., 0., 0., 0.],
             ...,
             [1., 0., 0., ..., 0., 0., 0.],
             [1., 0., 1., ..., 0., 0., 0.],
             [0., 1., 0., ..., 0., 0., 0.]])
```

Fig 3.2 - Diagram showing transformed features.

The x_reshaped array was renamed x_resampled and important libraries were imported for the Classification model.

Creating statistical model with RandomForestClassifier

```
[23]: from sklearn.ensemble import RandomForestClassifier
      from sklearn.metrics import accuracy_score, f1_score, precision_score
      from sklearn.model_selection import train_test_split
      from sklearn.svm import SVC
      from sklearn.linear_model import LogisticRegression
```

```
[24]: X_resampled
```

```
[24]: array([[0., 1., 1., ..., 0., 0., 0.],
             [1., 0., 1., ..., 0., 0., 0.],
             [1., 0., 1., ..., 0., 0., 0.],
             ...,
             [1., 0., 1., ..., 0., 0., 0.],
             [1., 0., 1., ..., 0., 0., 0.],
             [0., 1., 1., ..., 0., 0., 0.]])
```

```
[25]: X_train, X_test,y_train,y_test = train_test_split(X_resampled, y_resampled, test_size = 0.2, random_state = 42)
```

Fig 3.3 - Diagram showing X_resampled array.

- The model is trained using training to test-size of 80 to 20 ratio.
- Prediction is made on the test data.
- The accuracy score of the data is measured, generating a score of 0.84, which means our model is predicting well.

```
34]: X_train, X_test,y_train,y_test = train_test_split(X_resampled, y_resampled, test_size = 0.2, random_state = 42)
```

```
35]: model = RandomForestClassifier()
```

```
36]: model.fit(X_train,y_train)
```

```
36]: RandomForestClassifier
     RandomForestClassifier()
```

```
37]: y_pred = model.predict(X_test)
```

```
38]: accuracy = accuracy_score (y_pred, y_test)
     print (f'The accuracy score is {accuracy:.2f}')

     The accuracy score is 0.84
```

Fig 3.4 - Diagram showing the predictive model.